CDA 106
Numerical Methods for Data Science

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$$P_n(x) = f(x_0) + \frac{x-x_0}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0) + \cdots$$

$$+ \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{n!\,h^n}\Delta^n f(x_0)$$

Newton-Gregory forward difference Interpolation formula.

Consider, $\frac{x-x_0}{h} = q$ so that $\frac{x-x_1}{h} = q-1$ and so on

i.e. $\frac{x-x_{n-1}}{h} = q-(n-1)$

$\underbrace{x_0, x_1, x_2, \ldots, x_{n-1}, x_n}_{x \text{ any point}}$ ← $n+1$ points

$P_n(x) \simeq f(x)$

$$f(x) = f(x_0) + q\Delta f(x_0) + \frac{q(q-1)}{2!}\Delta^2 f(x_0) + \cdots$$

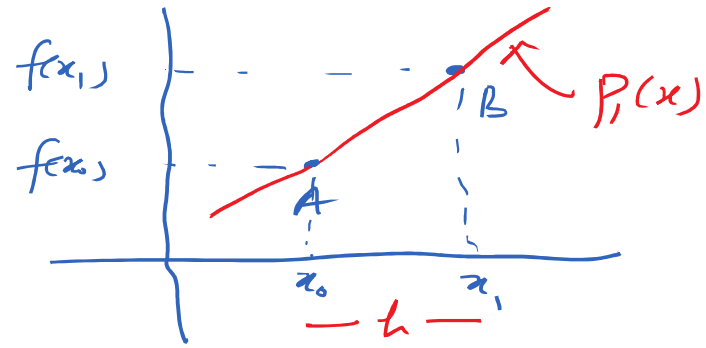
$$+ \frac{q(q-1)\cdots(q-(n-1))}{n!}\Delta^n f(x_0)$$

Newton's Interpolation

$A(x_0, f(x_0))$

$B(x_1, f(x_1))$

$$\boxed{p_1(x) = a_0 + a_1 x}$$



Expect to fit above data to a polynomial of degree 1

$f(x_0) = p_1(x_0)$, $f(x_1) = p_1(x_1)$

$p_1(x) = a_0 + a_1 x$ - - - (i)

$f(x_0) = a_0 + a_1 x_0$ - - - (ii)

$f(x_1) = a_0 + a_1 x_1$ - - - (iii)

Three ~~Two~~ eqn in two unknowns (a_0 & a_1)

Therefore, I can eliminate a_0 & a_1 from (i) – (iii)

On eliminating a_0 & a_1

$$\begin{vmatrix} p_1(x) & x & 1 \\ f(x_0) & x_0 & 1 \\ f(x_1) & x_1 & 1 \end{vmatrix} = 0$$

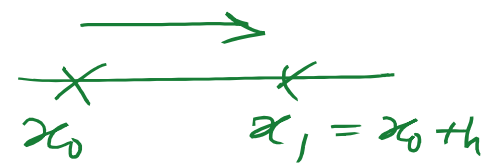
Expand the determinant along 1st row

$$P_1(x)\,(x_0 - x_1) - x\,(f(x_0) - f(x_1)) + 1\,(x_1 f(x_0) - x_0 f(x_1)) = 0 \quad (*)$$

Since, $x_1 - x_0 = h$ $\qquad \Delta f(x_0) = f(x_1) - f(x_0)$

$x_0 f(x_0) - x_0 f(x_0)$

$$\boxed{P_1(x) = f(x_0) + \frac{x - x_0}{h}\,\Delta f(x_0)}$$



Newton's Interpolation formula.

$$(*) \quad P_1(x)(-h) - x(-\Delta f(x_0)) + \frac{(x_1 - x_0)}{h} f(x_0) - x_0 \underbrace{(f(x_1) - f(x_0))}_{(\Delta f(x_0))} = 0$$

$$P_1 = \frac{1}{h}\left[h f(x_0) + \Delta f(x_0)(x - x_0)\right] = f(x_0) + \frac{x - x_0}{h}\,\Delta f(x_0)$$

Newton Gregory finite difference for n+1 points

Equidistant

$x_1 - x_0 = x_2 - x_1 = \cdots = x_n - x_{n-1} = h$

x_0	x_1	$\cdots$	x_n
$f(x_0)$	$f(x_1)$		$f(x_n)$

} n+1 data points

We want to fit an 'nth' degree polynomial "$p_n(x)$" such that

$$\boxed{p_n(x_i) = f(x_i) \ \forall i}$$

$$p_n(x) = a_0 + \underbrace{a_1(x-x_0)}_{1\text{ term}} + \underbrace{a_2(x-x_0)(x-x_1)}_{2\text{ terms}} + \cdots + \underbrace{a_n(x-x_0)(x-x_1)\cdots(x-x_{n-1})}_{n\text{ terms}} \quad (\#)$$

Therefore, at $x = x_0$

$p_n(x_0) = f(x_0)$

$(\#) \Rightarrow p_n(x_0) = f(x_0) = a_0 + 0 + 0 + \cdots + 0$

$\Rightarrow \boxed{a_0 = f(x_0)}$

Further

at $x = x_1$, $p_n(x_1) = f(x_1)$

$(\#) \Rightarrow f(x_1) = a_0 + a_1(x_1 - x_0) + 0 + \cdots + 0$

$\Rightarrow f(x_1) = f(x_0) + a_1 \cdot h$

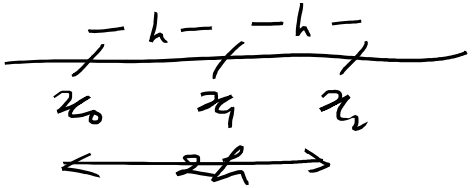
$\Rightarrow a_1 = \dfrac{f(x_1) - f(x_0)}{h} = \dfrac{\Delta f(x_0)}{h}$

at $x = x_2$

$(\#)$ $a_2 = \dfrac{\Delta^2 f(x_0)}{2!\, h^2}$

for $a_2 = ?$ put $x = x_2$ in (#) & note that $p_n(x_i) = f(x_i)$

$$(\#) \Rightarrow \quad f(x_2) = a_0 + a_1 \underbrace{(x_2 - x_0)}_{2h} + a_2 \underbrace{(x_2 - x_0)}_{2h} \underbrace{(x_2 - x_1)}_{h} + 0 \cdots + 0$$



$$\Rightarrow f(x_2) = f(x_0) + 2\cancel{h}\left[\frac{\Delta f(x_0)}{\cancel{h}}\right] + a_2 \cdot (2!\, h^2)$$

$$\Rightarrow f(x_2) - f(x_0) - 2\left(f(x_1) - f(x_0)\right) = a_2 (2!\, h^2)$$

$$\Rightarrow \underbrace{f(x_2) - 2f(x_1) + f(x_0)} = 2!\, h^2 \cdot a_2$$

$$\Rightarrow \quad a_2 = \frac{\Delta^2 f(x_0)}{2!\, h^2}$$

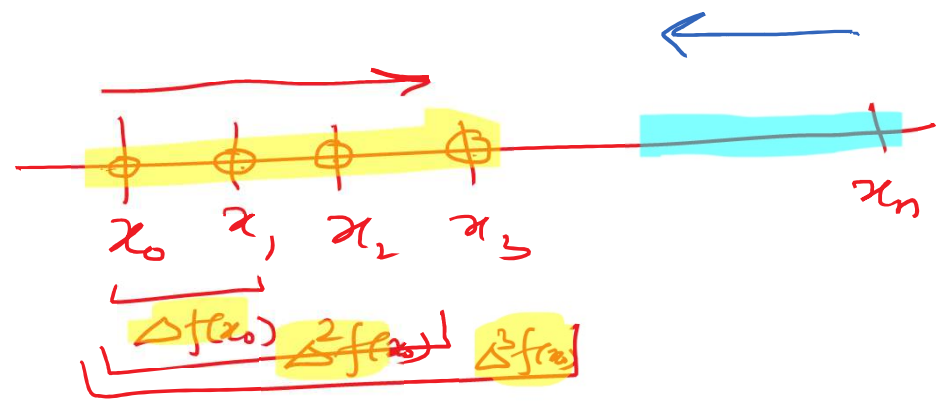
In the similar manner we can find

$$a_n = \frac{\Delta^n f(x_0)}{n!\, h^n}$$

Therefore, finally,

$$\boxed{P_n(x) = f(x_0) + \frac{(x-x_0)}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0) + \cdots + \frac{(x-x_0)\cdots(x-x_{n-1})}{n!\,h^n}\Delta^n f(x_0)}$$

Newton Gregory forward difference Interpolation formula.



The above formula is forward difference formula which is very useful when unknown is near the ~~beggin~~ beginning of the data.

However, when unknown is near the end of the data, then it is convenient to apply backward difference formula

Newton Gregory Backward difference Interpolation formula.

$\nabla \rightarrow$ Backward difference operator

$$\left(\frac{x - x_n}{h}\right) = q$$

$$f(x) = f(x_n) + q\,\nabla f(x_n) + \frac{q(q+1)}{2!}\nabla^2 f(x_n) + \frac{q(q+1)(q+2)}{3!}\nabla^3 f(x_n) + \cdots + \frac{q(q+1)\cdots(q+n-1)}{n!}\nabla^n f(x_n)$$

From Newton Gregory Backward Interpolation formula it can be easily seen.

$$f'(x) = \frac{df}{dq}\cdot\frac{dq}{dx} = \frac{1}{h}\left[\nabla f(x_n) + \frac{2q+1}{2!}\nabla^2 f(x_n) + \frac{3q^2+6q+2}{3!}\nabla^3 f(x_n) + \cdots\right]$$

&

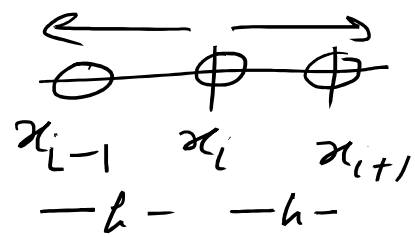
$$f''(x) = \frac{1}{h^2}\left[\nabla^2 f(x_n) + (q+1)\nabla^3 f(x_n) + \cdots\right]$$

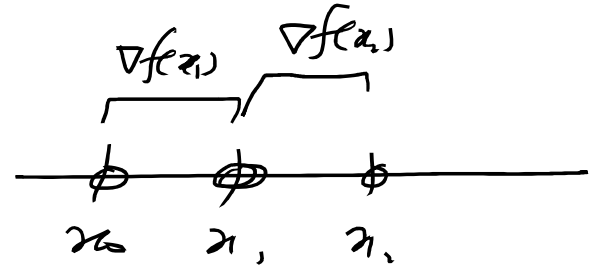
These formulae will be used to find derivatives at 'x'

Definition — Backward difference Operator (∇)

$$\boxed{\nabla f(x_i) = f(x_i) - f(x_{i-1})}$$

$$\boxed{\nabla f(x_i) = f(x_i) - f(x_i - h)}$$





for example

$$\nabla f(x_1) = f(x_1) - f(x_0) \quad \checkmark$$

$$\nabla f(x_2) = f(x_2) - f(x_1) \quad \checkmark$$

and

$$\nabla^2 f(x_2) = \nabla\big(\nabla f(x_2)\big) = \nabla\big(f(x_2) - f(x_1)\big)$$

$$= \nabla f(x_2) - \nabla f(x_1) = f(x_2) - 2f(x_1) + f(x_0)$$

In order to calculate $\nabla f(x_0), \nabla^2 f(x_0), \ldots$ etc. we use Backward difference Table

Ex

Data

x	1	2	3	4	5
$f(x)$	3	6	11	18	27

$h = 1$

x	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
x_0 1	3 $f(x_0)$			
		$\nabla f(x_1)$ 3 $= f(x_1) - f(x_0) = 6 - 3$		
x_1 2	6 $f(x_1)$		$\nabla^2 f(x_2)$ 2	
		$\nabla f(x_2)$ 5		0 $\nabla^3 f(x_3)$
x_2 3	11 $f(x_2)$		$\nabla^2 f(x_3)$ 2	
		$\nabla f(x_3)$ 7		0
x_3 4	18 $f(x_3)$		$\nabla^2 f(x_4)$ 2 $= \nabla(\nabla f(x_4))$	
		$\nabla f(x_4)$ 9		$\nabla^3 f(x_4)$
x_4 5	27 $f(x_4)$			

Q. Given the data in the previous slide fit a polynomial !

Ans Expect (4th degree or lower)

Apply Newton Gregory Backward Interpolation formula.

Note $q = \frac{x - x_4}{1} = (x-5)$

$x_4 = 5$, $\nabla f(x_4) = 9$, $\nabla^2 f(x_4) = 2$

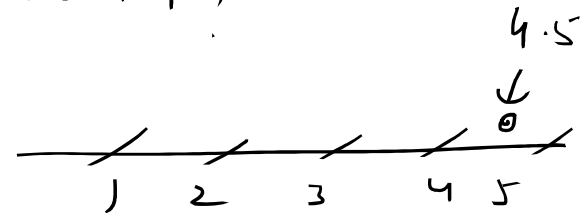
$f(x_4) = 27$, $\nabla^3 f(x_4) = 0$

$$f(x) = f(x_4) + q\nabla f(x_4) + \frac{q(q+1)}{2!}\nabla^2 f(x_4) + \frac{q(q+1)(q+2)}{3!}\cancelto{0}{\nabla^3 f(x_4)} + \xi_1 \xi \cancelto{0}{\nabla^4 f(x_4)}$$

$$= 27 + (x-5)9 + \frac{(x-5)(x-4)}{\cancel{2}} \times \cancel{2} + 0$$

$$= 27 + 9x - 45 + x^2 - 9x + 20 = x^2 + 2$$

Q. Further, find $f(4.5)$ from above data!



Note here $x = 4.5$ $\quad x_4 = 5$

$$q = \frac{4.5 - 5}{1} = -0.5$$

$$f(4.5) = f(x_4) + q\nabla f(x_4) + \frac{q(q+1)}{2}\nabla^2 f(x_4)$$

$$= 27 + (-0.5) \times 9 + \frac{(-0.5)(0.5)}{2} \times 2$$

$$= 27 - 4.5 - 0.25$$

$$f(4.5) = 22.25 \quad \text{Ans.}$$

Q. Suppose I want $f'(5) = ?$

Ans Note that $x = 5$ $\quad x_4 = 5$ $\quad q = \frac{x - x_4}{h} = \frac{5-5}{h} = 0$

Therefore, from formula.

$$f'(x) = \frac{1}{h}\left[\nabla f(x_4) + \frac{(2q+1)}{2!}\nabla^2 f(x_4) + \frac{3q^2+6q+2}{3!}\nabla^3 f(x_4) + \cdots\right]$$

on substitution

$$f'(5) = \frac{1}{1}\left[9 + \frac{2\times 0+1}{2!}\times 2 + 0\right] = 10$$

Ans

Exercise **Data**

x	1^{x_0}	2^{x_1}	3^{x_2}	4^{x_3}	5^{x_4}
$f(x)$	1	8	27	64	125

Q1 Fit the polynomial $(x^3 !!)$

Q2 find $f(4.5) = 91.125$

Q3 Find $f'(5) = 75$

Q4 find $f''(5) = 30$

Q5. find $f'(4.5) =$ left as exercise

(3×4.5^2).

$x_4 = 5$ $f(x_4) = 125$

$\nabla f(x_4) =$

$\nabla^2 f(x_4) =$

$\nabla^3 f(x_4) =$

$q =$?

Thank you!